

Title: AI Synthesis Documentation

Before

The original notebook contained working Python code, but variable names, formatting, and comments were limited. The code was functional but not fully organized according to PEP 8 guidelines.

Prompt

"Please reorganize my Python code following PEP 8 standards without changing the statistical analysis. Improve readability, add comments, and keep the outputs identical."

After

The code was reformatted with consistent indentation, descriptive comments, improved readability, and better organization while preserving the original statistical analyses, visualizations, and results.